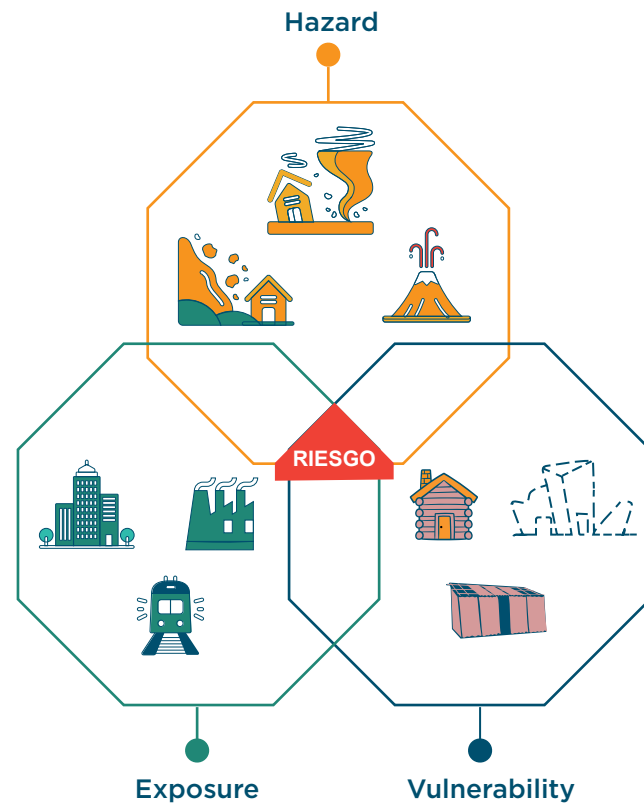




Disaster risk refers to the potential loss of life, injury, or destruction or damage as a result of the intersection of the hazard (natural hazard), the exposure of people or assets to the hazard, and the vulnerability (susceptibility to suffer damages) of the exposed assets to the hazard.



The Methodology consists of a series of phases and steps where efforts and resources are proportional to the risk level



Disaster and Climate Change Risk Assessment Methodology



Melissa Barandiarán
Maricarmen Esquivel
Sergio Lacambra Ginés
Suárez Daniela Zuloaga
Serge Troch

Where does Climate Change fit into disaster risk?

The intersection of disaster risk and climate change adaptation has been recognized by both the Intergovernmental Panel on Climate Change (IPCC) and the United Nations Office for Disaster Risk Reduction (UNDRR) through the Special Report on Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation (SREX). The influence of climate change on disaster risk translates mainly into adding a component of change and variability (and uncertainty) to the otherwise stationary treatment of hydrometeorological-related hazards (in the future) in disaster risk. In a way, climate change may be considered a disaster risk modification—and possibly exacerbation—factor.



Copyright © 2025 Inter-American Development Bank ("IDB"). This work is subject to a Creative Commons license CC BY 3.0 IGO (<https://creativecommons.org/licenses/by/3.0/igo/legalcode>). The terms and conditions indicated in the URL link must be met and the respective recognition must be granted to the IDB.

Further to section 8 of the above license, any mediation relating to disputes arising under such license shall be conducted in accordance with the WIPO Mediation Rules. Any dispute related to the use of the works of the IDB that cannot be settled amicably shall be submitted to arbitration pursuant to the United Nations Commission on International Trade Law (UNCITRAL) rules. The use of the IDB's name for any purpose other than for attribution, and the use of IDB's logo shall be subject to a separate written license agreement between the IDB and the user and is not authorized as part of this license.

Note that the URL link includes terms and conditions that are an integral part of this license.

The opinions expressed in this work are those of the authors and do not necessarily reflect the views of the Inter-American Development Bank, its Board of Directors, or the countries they represent.





Why should we care about disaster and climate change risk?



How to conduct a disaster and climate change risk screening?



What are the next steps after identifying risks?



How to incorporate disaster and climate change risk throughout the project cycle?

METHODOLOGY

Phase

1 Screening & Classification

Step 1

Hazard exposure

Preliminary classification based on location and hazards

Step 2

Criticality & vulnerability

Revision of classification based on criticality & vulnerability



No action necessary

2 Qualitative Assessment

Step 3

Narrativa

Simplified qualitative risk assessment (narrative with diagnostic) & management plan



Moderate

Existing measures are sufficient

Qualitative DRA & DRMP

Step 4

Qualitative analysis

Complete qualitative risk assessment & management plan



Moderate & High

Tolerable uncertainty And/or impacts

Qualitative DRA & DRMP

3 Quantitative Assessment

Step 5

Quantitative analysis

Quantitative risk assessment (scientific analysis quantifying risk) & management plan



Moderate & High

Tolerable risk

Quantitative DRA & DRMP

Identification »

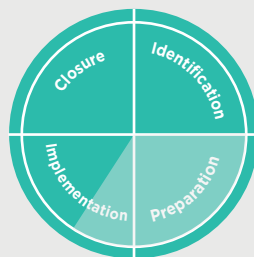
Preparation »

DRA: Disaster Risk Assessment
DRMP: Disaster Risk Management Plan

IMPLEMENTATION

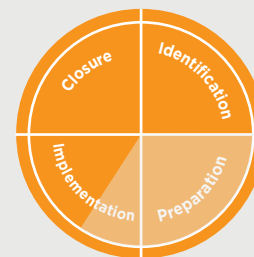
(Given that there are no life-cycle constraints, these steps may be completed after Approval)

The Methodology composed of 3 Phases (5 Steps) provides a clear and practical framework for the adequate incorporation of these risks in projects in a gradual and scalable manner.



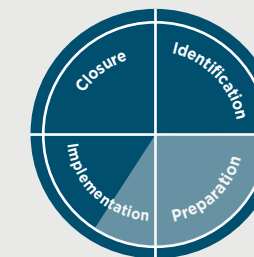
Phase 1

Applies to all projects. This Phase should be conducted as early as possible in the project cycle, and no later than the Project Profile, when there is a minimum definition of the project



Phase 2

Applies to all moderate or high risk projects. Some moderate risk projects may exit the process after conducting Step 3 if the analysis concludes that existing measures are sufficient. Some high risk projects may exit the process after conducting Step 4 if the analysis concludes that the uncertainties are tolerable and can be managed with the proposed measures. This Phase should be conducted once there is more definition of the project; Step 3 should be conducted before Approval.



Phase 3

Applies to the moderate or high risk projects that require a quantitative assessment according to the results of Phase 2. This Phase should be conducted once more detailed information is available, usually at the time of pre-feasibility designs.